

Assignment 3

1. a) absolute error, $E = 2 \text{ mph}$
 true value, $x = 60 \text{ mph}$

To find approximated value, $\bar{x}$:

$$E = |x - \bar{x}|$$

$$2 = |60 - \bar{x}|$$

$$60 - \bar{x} = 2$$

$$\bar{x} = 58 \text{ mph}$$

or

$$E_{rel} = \left| \frac{E}{x} \right|$$

$$= \left| \frac{2}{60} \right|$$

$$= 0.0333 \approx 3.33\%$$

$$60 - \bar{x} = -2$$

$$\bar{x} = 62 \text{ mph}$$

relative error, E_{rel} :

$$E_{rel} = \left| \frac{x - \bar{x}}{x} \right|$$

$$E_{rel} = \left| \frac{60 - 58}{60} \right|$$

$$= 0.0333$$

$$= 3.33\%$$

$$E_{rel} = \left| \frac{60 - 62}{60} \right|$$

$$= 0.0333$$

$$= 3.33\%$$

$\therefore$ relative error of the measurement is 3.33%

b) $f'(x) \approx \frac{f(x+h) - f(x)}{h}$

$h = 0.1$; $f(x) = x$

$$f'(2) = \frac{f(2+0.1) - f(2)}{0.1}$$

$$= \frac{f(2.1) - 2}{0.1}$$

$$= \frac{2.1 - 2}{0.1}$$

$$= \frac{0.1}{0.1}$$

$$= 1$$

absolute error, $E = |1 - 1|$
 $= 0$

$\therefore$ Absolute error for $f'(2)$ is 0

c) $6.232 + 7.0321 + 5.43 - 3.385 = 15.3091$

$$0.0005 + 0.00005 + 0.005 + 0.0005 = 0.00605$$

$$\text{Max} : 15.3091 + 0.00605 = 15.31515$$

$$\text{Min} : 15.3091 - 0.00605 = 15.30305$$

$$\therefore 15.3$$

2. $f(x) = x^3 - 6x + 1$

$$f(1.22) = 1.22^3 - 6(1.22) + 1$$

$$= -4.504152$$

$$\alpha = -4.504152$$

$$\bar{\alpha} = -4.504$$

Absolute error, $\epsilon = |\alpha - \bar{\alpha}|$

$$= |-4.504152 - (-4.504)|$$

$$= |-0.000152|$$

$$= 0.000152$$

Relative error, $\epsilon_{rel} = \left| \frac{\alpha - \bar{\alpha}}{\alpha} \right|$

$$= \left| \frac{-4.504152 - (-4.504)}{-4.504152} \right|$$

$$= 3.374664088 \times 10^{-5}$$

3. $f(x) = 3x - e^{-x}$

a) i) $[0.1, 1.5]$

$$f(0.1) = 3(0.1) - e^{-0.1}$$

$$= -0.6048$$

$$f(1.5) = 3(1.5) - e^{-1.5}$$

$$= 4.2769$$

$$f(0.1)f(1.5) = (-0.6048)(4.2769)$$

$$= -2.5867 (< 0)$$

$\therefore$ The root exist in the interval $[0.1, 1.5]$.

ii) $[1, 2]$

$$f(1) = 3(1) - e^{-1}$$

$$= 2.6321$$

$$f(2) = 3(2) - e^{-2}$$

$$= 5.8647$$

$$f(1)f(2) = (2.6321)(5.8647)$$

$$= 15.4365 (> 0)$$

$\therefore$ There is no root exist in the interval $[1, 2]$.

3. $f(x) = 3x - e^{-x}$

b) interval : $[0.1, 1.5]$

$E = 0.0001$

Bisection method

i	a	b	f(a)	f(b)	c	f(c)
0	0.1	1.5	-0.6048	4.2769	0.8	1.9507
1	0.1	0.8	-0.6048	1.9507	0.45	0.7124
2	0.1	0.45	-0.6048	0.7124	0.275	0.0654
3	0.1	0.275	-0.6048	0.0654	0.1875	-0.2665
4	0.1875	0.275	-0.2665	0.0654	0.2313	-0.0998
5	0.2313	0.275	-0.0998	0.0654	0.2532	-0.0169
6	0.2532	0.275	-0.0169	0.0654	0.2641	0.0244
7	0.2532	0.2641	-0.0169	0.0244	0.2587	0.0039
8	0.2532	0.2587	-0.0169	0.0039	0.2560	-0.0063
9	0.2560	0.2587	-0.0063	0.0039	0.2574	-0.0010
10	0.2574	0.2587	-0.0010	0.0039	0.2581	-0.0016
11	0.2574	0.2581	-0.0010	0.0016	0.2578	0.0005
12	0.2574	0.2578	-0.0010	0.0005	0.2576	-0.0001
13	0.2576	0.2578	-0.0001	0.0005	0.2577	0.0003
14	0.2576	0.2577	-0.0001	0.0003	0.2577	0.00008

$\therefore$ In 14th iteration, the value $f(c) = 0.00008 < 0.0001$

Hence, the root of $f(x)$ in interval $[0.1, 1.5]$ is 0.258

4. $f(x) = \cos(x) + 2 \sin(x) + x^2$

a) Secant method

$$x_0 = 0 ; x_1 = -0.1$$

$$e = \frac{1}{2} \times 10^{-4}$$

$$= 0.00005$$

$$f(0) = \cos 0 + 2 \sin 0 + 0^2$$

$$= 1$$

$$f(-0.1) = \cos(-0.1) + 2 \sin(-0.1) + (-0.1)^2$$

$$= 0.8053$$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$= \frac{(0)(0.8053) - (-0.1)(1)}{0.8053 - 1}$$

$$= -0.5136$$

$$f(x_2) = \cos(-0.5136) + 2 \sin(-0.5136) + (-0.5136)^2$$

$$= 0.1521$$

i	x_i	x_{i+1}	x_{i+2}	$f(x_{i+2})$
0	0	-0.1	-0.5136	0.1521
1	-0.1	-0.5136	-0.6099	0.0461
2	-0.5136	-0.6099	-0.6518	0.0067
3	-0.6099	-0.6518	-0.6589	0.0003
4	-0.6518	-0.6589	-0.6592	0.00003

$\therefore$ The root of $f(x)$ is $-0.6592 \approx -0.66$

4. $f(x) = \cos(x) + 2\sin(x) + x^2$

b) Newton method

$$x_0 = -0.1$$

$$\epsilon = 0.00005$$

$$\begin{aligned} f(-0.1) &= \cos(-0.1) + 2\sin(-0.1) + (-0.1)^2 \\ &= 0.8053 \end{aligned}$$

$$f'(x) = -\sin(x) + 2\cos(x) + 2x$$

$$\begin{aligned} f'(-0.1) &= -\sin(-0.1) + 2\cos(-0.1) + 2(-0.1) \\ &= 1.8898 \end{aligned}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_1 = -0.1 - \frac{0.8053}{1.8898}$$

$$= -0.5261$$

n	x_n	$f(x_n)$	$f'(x_n)$
0	-0.1	0.8053	1.8898
1	-0.5261	0.1372	1.1795
2	-0.6424	0.0151	0.9156
3	-0.6589	0.0003	0.8758
4	-0.6592	0.00006	0.8750
5	-0.6593	-0.000002	0.8748
6	-0.6593		

$\therefore$ the root is $-0.6593 \approx -0.66$